

Part One

University of Birmingham
Geometry and Mathematical Physics Seminar

- Axiomatic Approaches to Quantum Field Theory.
- Functorial Field Theory
- von Neumann Algebras

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§. Introduction

Hilbert 1901 List of 23 Problems

Sixth Problem

Formalisation / Axiomatisation of Physics

Mathematical Treatment of the Axioms of Physics.

Axiomatic Approaches to Quantum Field Theory

Wightman Axioms
1960s

Fields

'operator valued distributions'

Haag Kastler Axioms
1960s

Algebra of Observables

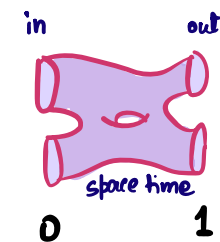


AQFT

Algebraic QFT

Atiyah-Segal formalism
late 1980s

Space of States and propagators.



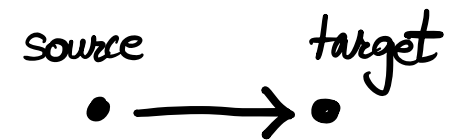
$H_0 \xrightarrow{f} H_1$

FQFT

Functorial QFT

E. Functorial Approach to Quantum Field Theory,

Categories and Functors



A category consists of a collection of objects and a collection of morphisms

every morphism has a source object and a target object

we can compose morphisms, there is a unique identity morphism for each object.

Composition is associative, satisfies left and right unit laws.

Examples (objs and morphisms)

Set (sets and functions)

$\text{Vect}_{\mathbb{C}}$ (vector spaces and linear maps)

Grp (groups and group homomorphisms)

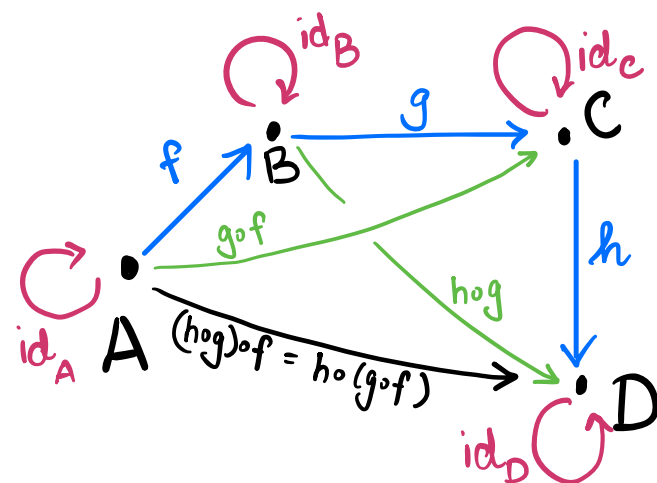
Top (topological spaces and cont. maps)

Cat (categories and functors)

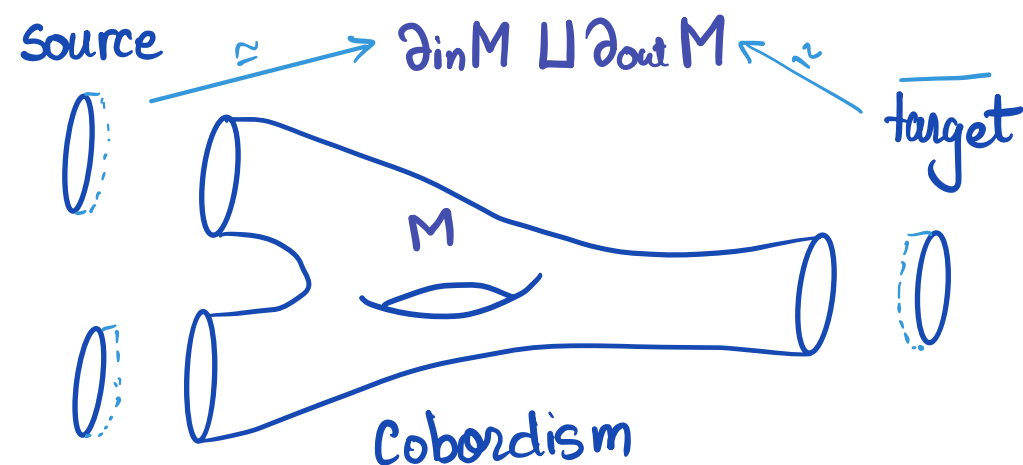
$\text{Cob}_{n-1,n}$ (closed $n-1$ dim manifolds and n -dim cobordisms)

upto diffeomorphisms.

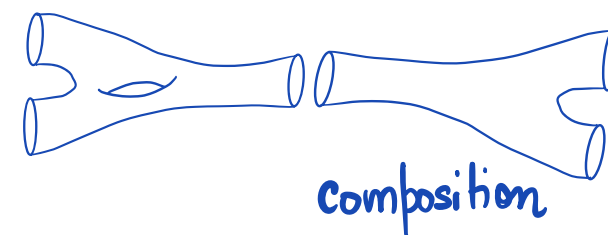
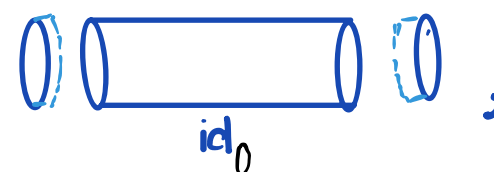
category



$\partial M =$



$\text{Cob}_{1,2}$



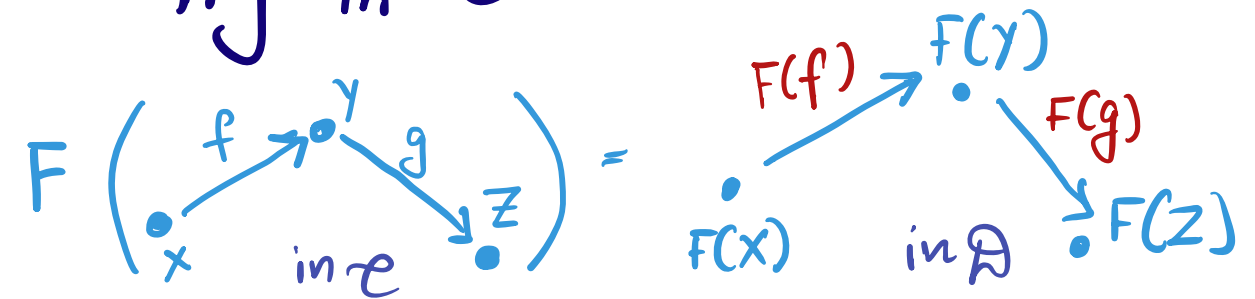
composition



gluing

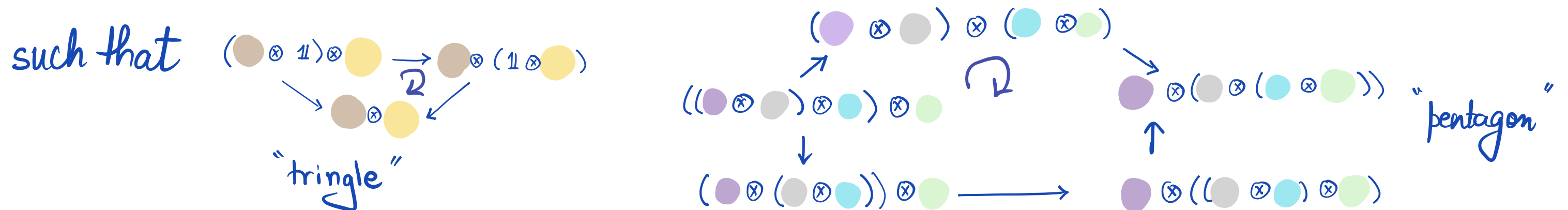
Cobordism

A **functor** $F: \mathcal{C} \rightarrow \mathcal{D}$ is a map sending each object C in $\mathcal{C}$ to an object $F(C)$ in $\mathcal{D}$ and each morphism $f: X \rightarrow Y$ in $\mathcal{C}$ to a morphism $F(f): F(X) \rightarrow F(Y)$ such that $F(g \circ f) = F(g) \circ F(f)$ for all composable pairs f, g in $\mathcal{C}$ and $F(\text{id}_C) = \text{id}_{F(C)}$ for all objects in $\mathcal{C}$.



A **monoidal category** is equipped with a notion of product of its objects (and morphisms)

- * a functor $\otimes: \mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$
- * a unit object $1 \in \mathcal{C}$
- * associators $(_ \otimes _) \otimes _ \xrightarrow{\cong} _ \otimes (_ \otimes _)$
- * left and right unitors $1 \otimes _ \xrightarrow{\cong} _, _ \otimes 1 \xrightarrow{\cong} _$

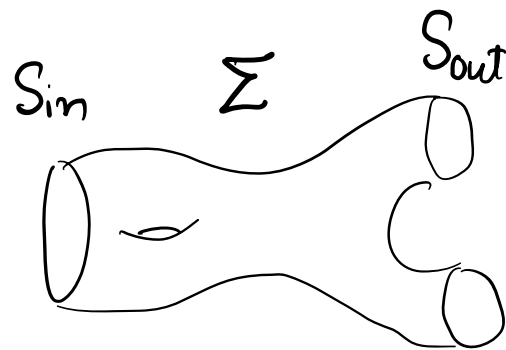


Axiomatise the properties of a Path Integral

A d -dim functorial field theory is a symmetric monoidal functor

$$Z: \text{Cob}_{d, d-1}^{\mathcal{G}} \longrightarrow \text{Vect}_{\mathbb{C}}$$

closed $d-1$ manifold $S \longmapsto Z(S)$
the vector space corresponding to space of states.



$\longmapsto Z(\Sigma): Z(S_{in}) \rightarrow Z(S_{out})$
a linear map (propagator)

monoidality
? Quantum.

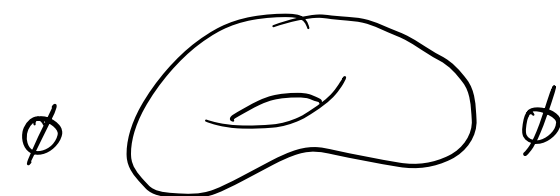
$$\begin{array}{c} \sqcup \\ \emptyset \end{array}$$

$$\longmapsto$$

$$\otimes$$

$$\longmapsto$$

$$\mathbb{C}$$



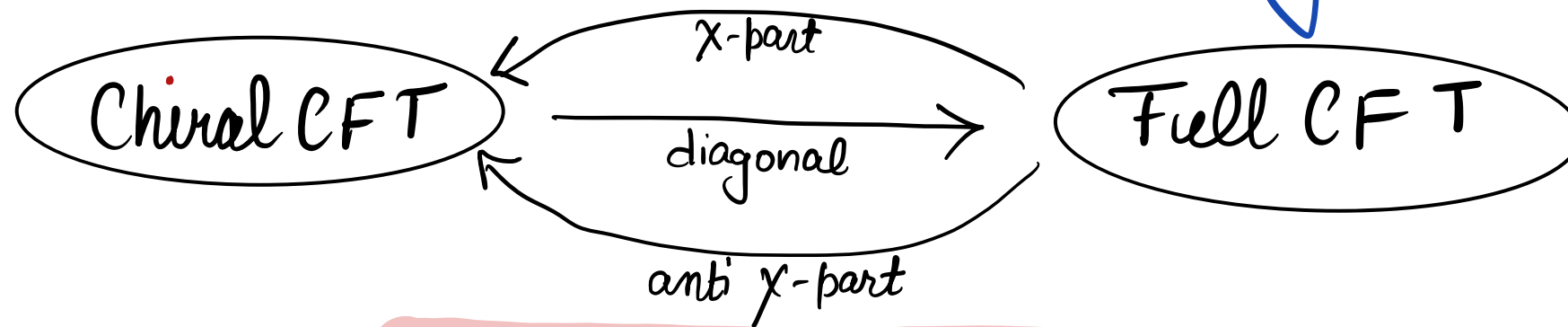
closed d -manifold

$$\longmapsto$$

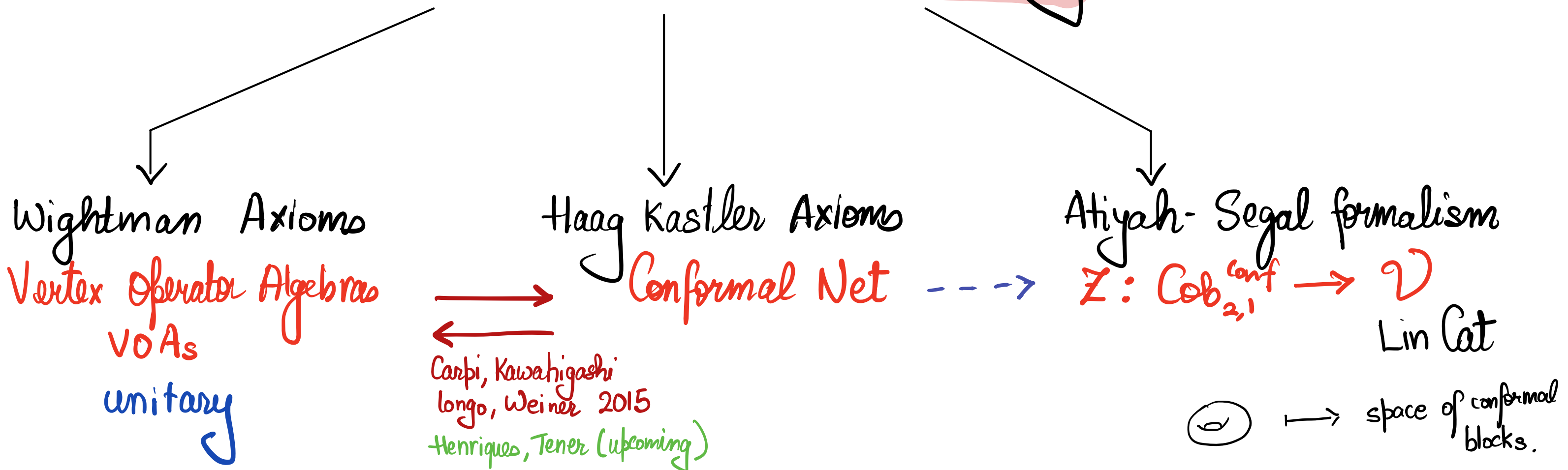
$$\mathbb{C} \longrightarrow \mathbb{C} \quad \epsilon \in \mathbb{C}$$

the partition function

2d Conformal Field Theory



2d Chiral Conformal field theory



2d chiral conformal field theory sits at the boundary of 3d TQFT

Review of vN algebras

Ref. Complete W^* -categories (N-Henriques - Penneys)

Let H be a Hilbert space.

let $B(H)$ be the $*$ -algebra of bounded linear operators on H .

Defn.

A von Neumann algebra on H is a $*$ -subalgebra of $B(H)$ such that $A'' = A$ where $A' = \{b \in B(H) ; ab = ba \ \forall a \in A\}$ and $A'' := (A')'$.

A left module over a von Neumann algebra A is a Hilbert space ${}_A H$ equipped with a homomorphism $A \rightarrow B(H)$.

An A - B -bimodule is a Hilbert space ${}_A H_B$ equipped with $A \rightarrow B(H)$, $B^{op} \rightarrow B(H)$ whose images commute in $B(H)$

An abstract von Neumann algebra (W^* -algebra) is a $*$ -algebra isomorphic to a von Neumann algebra on some Hilbert space.

By Haagerup 75, every abstract vN algebra A admits a canonically associated Hilbert space ${}_A L^2 A_A$ which is an A - A bimodule.

This is called its standard form (non-commutative L^2 -space).

Let $\lambda: A \rightarrow \mathcal{B}(L^2 A)$ and $\rho: A^{\text{op}} \rightarrow \mathcal{B}(L^2 A)$ be the actions.

Defn. Let H_A and ${}_A K$ be right and left A -modules respectively.
 Their Connes fusion $H \boxtimes_A K$ is the Hilbert space completion

of the vector space $\underbrace{\text{Hom}_{\text{Mod-}A}({}_A L^2 A, H_A)}_{\otimes_A} \underbrace{{}_A L^2 A}_A \otimes_A \underbrace{\text{Hom}_{A\text{-Mod}}({}_A L^2 A, {}_A K)}_{\otimes_A}$

with respect to the inner product

$$\langle \varphi_1 \otimes \xi_1 \otimes \psi_1, \varphi_2 \otimes \xi_2 \otimes \psi_2 \rangle := \langle \xi_1, \lambda^{-1}(\varphi_1^* \circ \varphi_2) \xi_2 \rho^{-1}(\psi_1^* \circ \psi_2) \rangle_{L^2 A}$$

$\boxtimes_A$ is associative and $L^2 A$ is the unit.

Alternative expression

$$H \boxtimes_A K = \text{proj}_{\bar{H}} L^2 \left(\text{End}_{A\text{-Mod}} ({}_A \bar{H} \oplus {}_A K) \right) \text{proj}_K$$

$$\text{proj}_{2,1} \rightarrow \left(\begin{array}{c} \boxed{} \end{array} \right)$$

Next : partial construction of
Segal (functorial) χ -CFTs from Conformal Nets.

Part Two

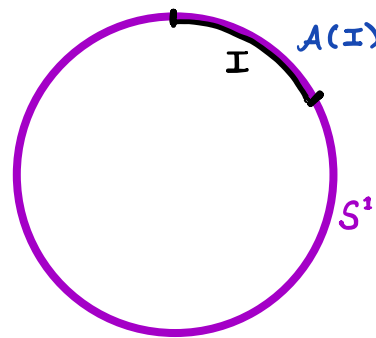
§. Conformal Nets

§. Solitonic Representations of Conformal Nets

§. Role of 2-Hilb and 3-Hilb (higher hilbert spaces)

§. Functorial assignment to cobordisms
(0 and 1 dimensions).

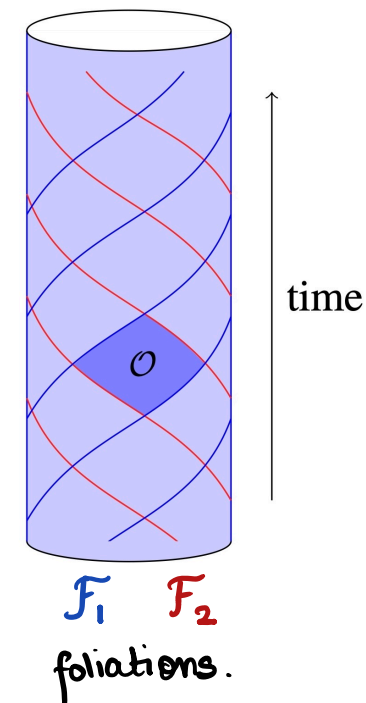
Conformal Nets



vacuum sector We define a conformal net on S^1 to be the data of a Hilbert space H_0 equipped with a projective unitary representation U of $\text{Diff}^+(S^1)$ on H_0 and a pointing $\Omega \in H_0$ called the vacuum vector, as well as the data of a von Neumann algebra $\mathcal{A}(I) \subseteq B(H_0)$ for every subinterval $I \subsetneq S^1$. These are required to satisfy the following properties:

- **Isotony:** For any two subintervals $I, I' \subset S^1$, if $I \subseteq I'$, then $\mathcal{A}(I) \subseteq \mathcal{A}(I')$.
- **Locality:** For any two subintervals $I, I' \subset S^1$, if $\text{int}(I) \cap \text{int}(I') = \emptyset$, then $[\mathcal{A}(I), \mathcal{A}(I')] = 0$.
 $B(H_0)$
- **Covariance:** For any $g \in \text{Diff}^+(S^1)$ and any interval $I \subset S^1$, we have $\text{Ad}_{U(g)}(\mathcal{A}(I)) = \mathcal{A}(g(I))$.
- **Vacuum** For any $g \in \text{Möb}$, Ω is preserved by $U(g)$ (upto a non-zero scalar multiple) and Ω is cyclic for $\bigvee_{I \subset S^1} \mathcal{A}(I)$.
- **Positivity of the energy** The U is a positive energy representation, that is, the generator of the one-parameter rotation subgroup has positive spectrum.

M :



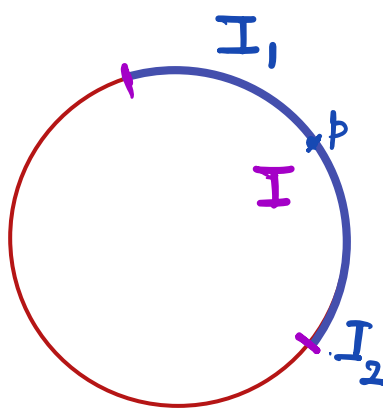
Chiral: $\mathcal{A}(O)$ only depends on image of O under projection $M \rightarrow M/F_1$

Introduced in Gabbiani-Frölich '93.

The following properties hold for any conformal net.

- **Reeh-Schlieder property** The vacuum vector Ω is cyclic and separating for each local algebra $\mathcal{A}(I)$ for each $I \subsetneq S^1$.
- **Bisognano-Wichmann property** For every interval $I \subsetneq S^1$, there exists a one-parameter subgroup $\Lambda_I(t) \subset \text{Möb}$ which fixes the boundary points of I , such that $U(\Lambda_I(2\pi t)) = \Delta_I^{it}$ where Δ_I is the modular operator associated to $\mathcal{A}(I)$ and $\Omega \in H_0$. **(Tomita-Takesaki)**
- **Haag duality** The commutant of $\mathcal{A}(I)$ in $B(H_0)$ is $\mathcal{A}(S^1 \setminus \text{int}(I))$.
- **Factoriality** Each local algebra $\mathcal{A}(I)$ is a type III₁ factor. *trivial center*
- **Additivity** Let $\mathcal{I}$ denote a covering of the interval I , then $\mathcal{A}(I) \subset \bigvee_{J \in \mathcal{I}} \mathcal{A}(J)$.

For some conformal nets, a property called *strong additivity* holds. A conformal net $\mathcal{A}$ is strongly additive if $\mathcal{A}(I_1) \vee \mathcal{A}(I_2) = \mathcal{A}(I)$ whenever I_1 and I_2 are the closures of connected components of $I \setminus \{p\}$ for an interior point p interval I . In the following work, we relax this condition and hence we also do not limit ourselves to rational conformal nets. Virasoro nets with $c > 1$ fail to have strong additivity.



Thm. From the data of a conformal net A on S^1 ,
we can build a co-ordinate free conformal net,

a covariant functor

$$A : \begin{array}{c} \text{Int} \\ \text{intervals} \\ \text{smooth embeddings} \end{array} \longrightarrow \begin{array}{c} \text{vNAlg} \\ \text{von Neumann algebras} \\ \text{vNalg. homomorphisms} \end{array}$$

A representation of a conformal net A on a circle S is a Hilbert space H equipped with actions $\alpha_I: A(I) \rightarrow B(H)$ $\forall I \subsetneq S$ such that

$$J \subset I \Rightarrow \alpha_I|_{A(J)} = \alpha_J.$$

A morphism between two representations of A

$$f: (H_1, \{\alpha_I\}_{I \subsetneq S}) \rightarrow (H_2, \{\beta_I\}_{I \subsetneq S})$$

is a map $f: H_1 \rightarrow H_2$ of Hilbert spaces such that $f \circ \alpha_I = \beta_I \quad \forall I \subsetneq S$.

The category of representations of A on circle S is denoted $\text{Rep}_S(A)$

Given a diffeomorphism $\phi: S_1 \rightarrow S_2$, there is an equivalence of categories

$$\phi^*: \text{Rep}_{S_2}(A) \rightarrow \text{Rep}_{S_1}(A)$$

$$(H, \{\alpha_{I_2}\}_{I_2 \subsetneq S_2}) \mapsto (H, \{\alpha_{\phi(I_1)} \circ A(\phi|_{I_1})\}_{I_1 \subsetneq S_1})$$

non-degenerate ribbon finite tensor category

For rational CFTs,
 $\text{Rep}_s(A)$ is a modular tensor category (MTC).

} Reshetikin-Turaev construction
↓

3D TQFT " $Z: \text{Cob}_{2,3} \rightarrow \text{Vect}$ " (modulo anomaly)

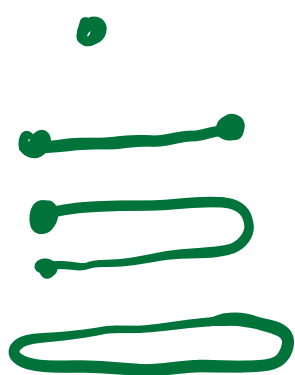
Reshetikin-Turaev state space on a 2-manifold Y
 $\simeq$ space of conformal blocks on Y (holographic principle)

What does this assign to a point? Cobordism Hypothesis for TQFTs.

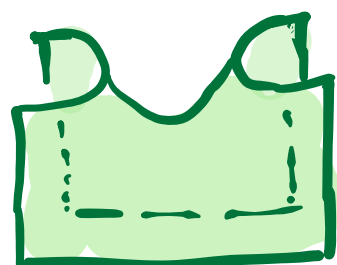
Fully - Extended functorial field theories

$$Z : \text{Cob}_3 \rightarrow \mathcal{V}$$

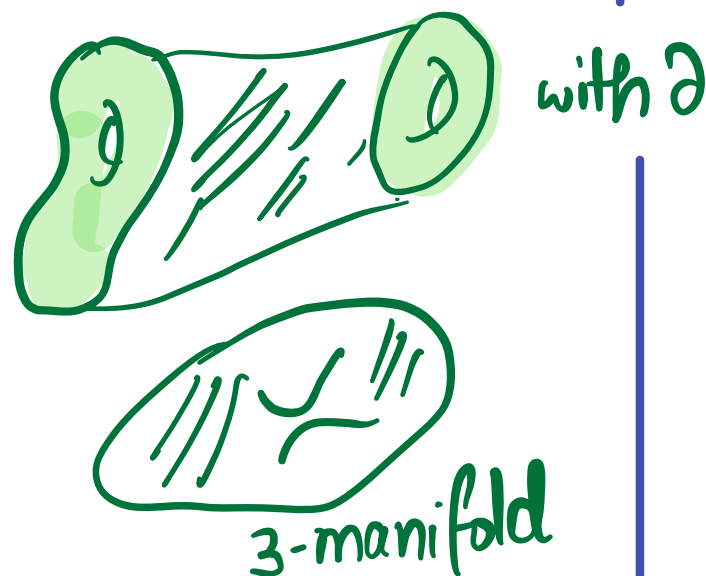
obj :
1-mor :



2-mor :



3-mor :



with ∂

3-manifold

Algebraic 3-category

monoidal unit.

$$\text{End}_{n\text{-Hilb}}(\mathbb{1}) \approx (n-1)\text{-Hilb}$$

$$\Omega_1 n\text{-Hilb} \text{ (loops)}$$

Model for 2-Hilb
Mor(VN Alg)

von Neumann algebras
bi-modules
bi-module homomorphisms
monoidal unit $\mathbb{C}$

obj
1-mor
2-mor

Model for 3-Hilb

Mor (Categorifield VN algebras)

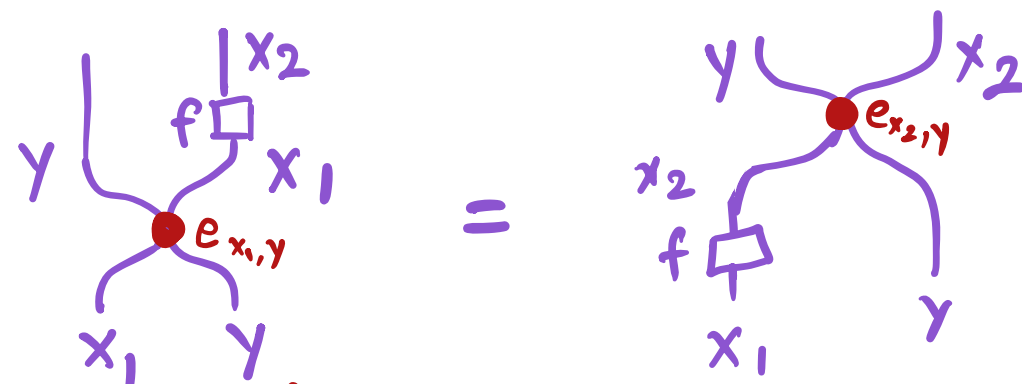
Bicommutant categories

Sketch of definition (analogous to vN algebras)

A
 algebra _{$\mathbb{C}$}
 $*$ -structure
 commutant A' in B
 given $A \hookrightarrow B$
 $\{b \in B : ab = ba \ \forall a \in A\}$
 * Commutant of a $*$ -alg
 is a $*$ -algebra.

$\mathcal{T}$
 $\mathbb{C}$ -linear monoidal category
 full-dagger (bi-involutive)-structure

given $i : \mathcal{T} \rightarrow \mathcal{E}$ a tensor functor
 $\mathcal{Z}(i : \mathcal{T} \rightarrow \mathcal{E})$ is the category
 obj: $(X, \{e_{x,y} : X \otimes i(Y) \xrightarrow{\sim} i(Y) \otimes X\}_{Y \in \mathcal{T}})$
 half-braidings natural in Y
 mor: $(X_1, \{e_{x_1}\}) \rightarrow (X_2, \{e_{x_2}\})$
 is $f : X_1 \rightarrow X_2$ in $\mathcal{E}$ such that



* has a monoidal structure.

example to keep in mind
 Hilb

(anti-linear anti tensor involution)

(anti-linear)

$${}_A L^2 A_A$$

$$\tau \tau^{\text{abs}} \tau$$

(full-subcategory generated by absorbing objects

$$\Omega \otimes X \cong \Omega \cong X \otimes \Omega \\ \forall X \in \tau.$$

$$L : \tau \rightarrow \text{End}(\tau^{\text{abs}}), \quad R : \tau^{\text{mop}} \rightarrow \text{End}(\tau^{\text{abs}}) \\ X \mapsto X \otimes _, \quad X \mapsto _ \otimes X$$

Bicommutant Category (abstract) is τ such that left and right actions of τ on τ^{abs} are each others commutants.

(data⁺ of L, R being bi-involutive tensor functors + lots of coherences ...)

Ex. Solitonic Representations of a conformal net.

Let A be a type III von Neumann algebra (factor)

$\text{End}(A)^{uo} \longrightarrow \text{Bim}(A)$ is monoidal equivalence.

obj $P: A \rightarrow A \longmapsto {}^P L^2 A \quad (A \xrightarrow{P} A \rightarrow \mathcal{B}(L^2 A))$

mor $b_{P_1, P_2}: P_1 \rightarrow P_2 \longmapsto {}^{P_1} L^2 A \rightarrow {}^{P_2} L^2 A$

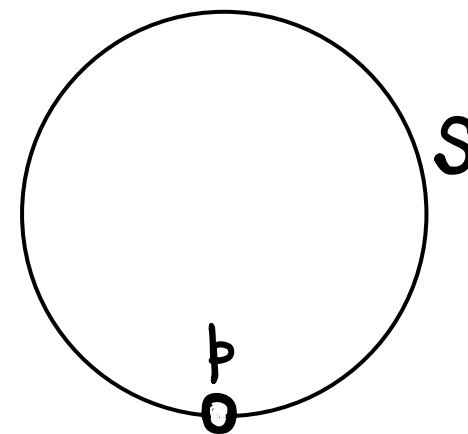
$b_{P_1, P_2} \in A$ s.t.
 $b_{P_1}(a) = P_2(a)b$
 $\forall a \in A$

act by b_{P_1, P_2}

$\otimes \quad P_1 \otimes P_2 := P_2 \circ P_1 \longmapsto {}^{P_1} L^2 A \boxtimes_A {}^{P_2} L^2 A \simeq {}^{P_2 \circ P_1} L^2 A$

Defn. Given a conformal net A , a circle S with a marked point p , the category of solitonic representations $\mathcal{T}_A(S, p)$ has

objects $(H, \{\alpha_J : A(J) \rightarrow B(H)\}_{J \subseteq S}^{\text{isotonic}})$
 $p \notin \overset{\circ}{J}$

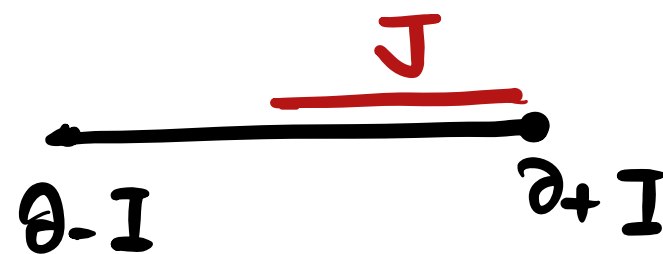


morphisms: maps of Hilbert spaces that intertwine all the actions

Defn. Given a conformal net A , an interval I , the category of left localised endomorphisms of $A(I)$, $\mathcal{T}_A^{\text{left}}(I)$

obj $P : A(I) \rightarrow A(I)$ such $\exists J \subset I, \partial_+ I \in J$
 $P|_{A(J)} = \text{id}_{A(J)}$

mor $a : P_1 \rightarrow P_2$ is an element in A (neighbourhood of $\partial_- I$) and $a P_1(x) = P_2(x) a \quad \forall x \in A(I)$.



Prop. : let $i: I \rightarrow I'$ be an embedding that maps $\partial I \rightarrow \partial I'$.
 $T_A^{\text{left}}(I)$ and $T_A^{\text{left}}(I')$ are monoidally equivalent.
 $p \mapsto p^{\text{ext}}$ (extension by identity)

Thm:
 (N.)

$$\begin{array}{ccc}
 T_A(S, p) & \longrightarrow & \text{Bim}(A(I)) \\
 \uparrow \cong & & \uparrow \cong \\
 T_A^{\text{left}}(I)^{\vee \otimes} & \longrightarrow & \text{End}(A(I))^{\vee \otimes}
 \end{array}$$

$\text{Sol}(A)$

Thm: $\text{Sol}(A)$ is a bicommutant category
 (N, Henriques)

Thm: Drinfeld's centre of $\text{Sol}(A)$ is $\text{Rep}(A)$
(N, Henriques)

Generalisation (work in progress)

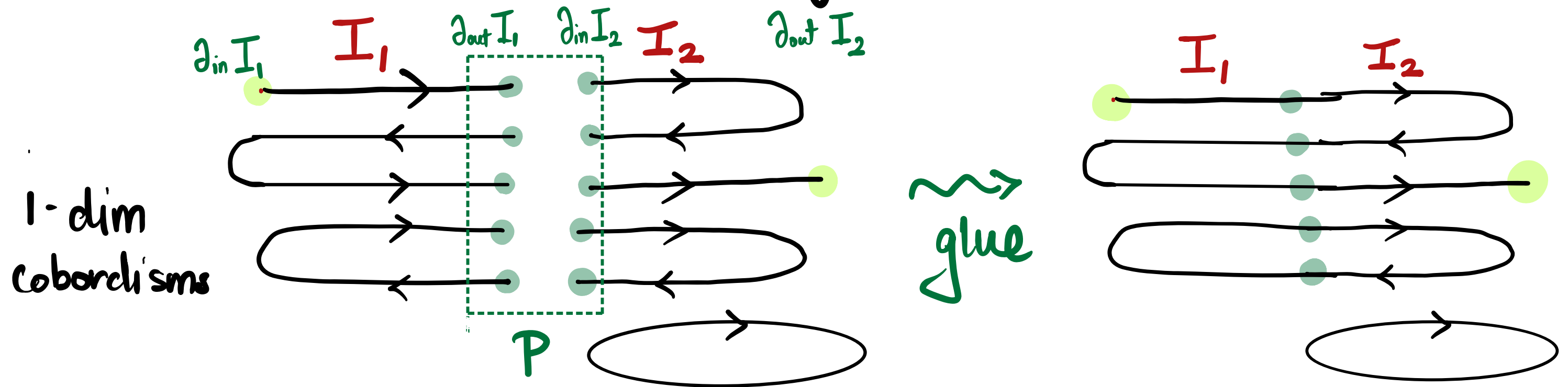
Given $B \subset A$ a subnet of a conformal net

we can define $\text{Sol}_B(A)$ of B -topological
solutions of A .

* Special case: $\text{Vir}_c \subset A$

we expect that $\text{Sol}_{\text{Vir}_c}(A)$ to be the category of
topological defects in 2D χ CFT A . 

Functorial assignment (dim 0 and 1)



The category of

$\partial_{in} I$ gets assigned

left-localised endomorphisms
of a collar neighbourhood of $\partial_{in} I$

$\partial_{out} I$ gets assigned

right-localised endomorphisms of
collar neighbourhood of $\partial_{out} I$

I gets assigned $\tau_{(\partial_{in} I)}^{\text{left}} A(I) - \text{Mod} \tau_{(\partial_{out} I)}^{\text{right}}$

Fusion (Categorified 'Connes fusion')

$$\star A(I_1) - \text{Mod} \cdot \boxed{\times} \cdot A(I_2) - \text{Mod} \cdot$$

$\tau_A(P)$

$$:= \text{Proj}_{2,1} \left(\text{End}_{\tau_A(P)} \left(\overline{\star A(I_1) - \text{Mod}} \oplus \star A(I_2) - \text{Mod} \right)^{\text{abs}} \right)$$

Thm (N.)

$$\star A(I_1 \cup_P I_2) - \text{Mod} \cdot$$

2)

Open : Physical interpretation of $\text{Sol}(A)$

To what extent does $\text{Sol}(A)$ remember the central charge of the theory?

[For rational theories, the drinfel'd center of $\text{Sol}(A)$ remembers $c \pmod{8}$]

Thank you

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